

Discrimination and Classification

Johnson, RA & Wichern, DW. "Applied Multivariate Analysis", Chapter 11, 6th edition, Pearson International Edition, 2007.

This chapter will focus on

(1) SEPARATING distinct sets of observations, and (2) ALLOCATING new observations to previously defined groups.

The two goals can be described as

DISCRIMINATION = finding "discriminants", i.e. features, whose numerical values are such that collections of observations are separated as much as possible.

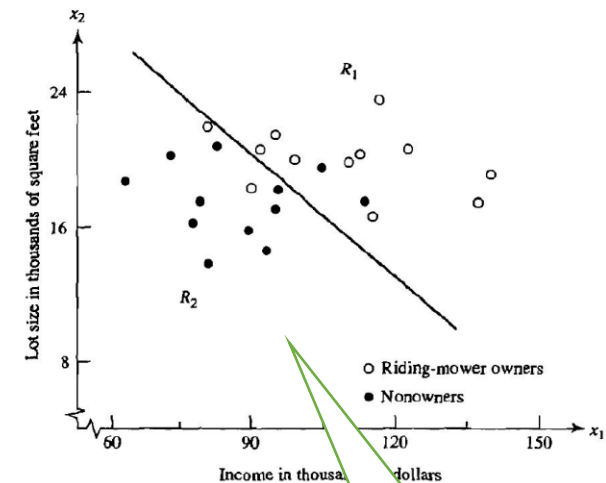
CLASSIFICATION = deriving a rule that can optimally allocate new observations to labelled classes.

Examples include

- Using measurements on education, income, family size, amount of previous brand switching to classify consumers as either "purchasers of a new product" or as "laggards".
- Using measurements on income, age, number of credit cards and family size to classify people as (1) good or (2) poor credits risks.
- Using measurements on easily observed external symptoms to classify people as either (1) diseased, or (2) nondiseased.

Contents:

- Introducing ideas of classification and misclassification
- EMC = Expected Cost of Misclassification
- Classification with two multivariate normal distributions
- Discrimination and classification from sample quantities
- Fisher's Discriminant Analysis
- Quadratic Discriminant Analysis
- Evaluating classification functions
- Classification of several populations



The figure above illustrates the separation (or overlap) of owners and non-owners of riding lawnmo-ers in terms of their income and the lot size of their property.

An "optimal" classification rule should

- result in few misclassifications;
- should take into account prior probabilities of occurrence of the two populations;
- take into consideration the cost of misclassification.

Can often be illustrated using a classification matrix:

Classification based on X	True group membership	
	π_1	π_2
$X \in R_1$	Correct classification	Classification error
$X \in R_2$	Classification error	Correct classification

Shows that owners and non-owners can be separated by linear line. Also shows that owners had higher income whilst separation based on lots size slightly less clear.

Let's introduce some notation:

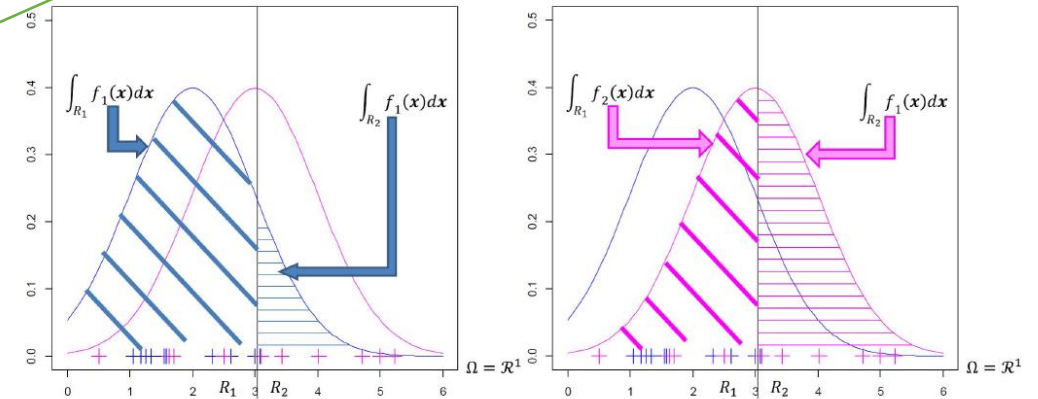
- π_1 and π_2 = the two labeled classes or populations.
- $\mathbf{X}' = [X_1, X_2, \dots, X_p]$ = the p random variables.
- $\mathbf{x}$ = the population values for $\mathbf{X}$ for populations π_1 and π_2 .
- $f_1(\mathbf{x})$ and $f_2(\mathbf{x})$ the probability density functions for the $\mathbf{x}$ values in the two populations.

The two probabilities of misclassification are

$$P(2|1) = P(\mathbf{X} \in R_2 | \pi_1) = \int_{R_2} f_1(\mathbf{x}) d\mathbf{x},$$

and

$$P(1|2) = P(\mathbf{X} \in R_1 | \pi_2) = \int_{R_1} f_2(\mathbf{x}) d\mathbf{x}.$$



This integral measure the area under $f_1(\mathbf{x})$ that falls in region 2. So $f_1(\mathbf{x})$ refers to correct population and integral measures misclassifications.

If p_1 is the prior probability of π_1 and p_2 is the prior probability of π_2 with $p_1 + p_2 = 1$ then

$$P(\text{correct classification as } \pi_1) = P(\mathbf{X} \in R_1 \cap \pi_1) = P(\mathbf{X} \in R_1 | \pi_1)P(\pi_1) = p_1 \int_{R_1} f_1(\mathbf{x}) d\mathbf{x}$$

$$P(\text{misclassification as } \pi_1) = P(\mathbf{X} \in R_1 \cap \pi_2) = P(\mathbf{X} \in R_1 | \pi_2)P(\pi_2) = p_2 \int_{R_1} f_2(\mathbf{x}) d\mathbf{x}$$

$$P(\text{correct classification as } \pi_2) = P(\mathbf{X} \in R_2 \cap \pi_2) = P(\mathbf{X} \in R_2 | \pi_2)P(\pi_2) = p_2 \int_{R_2} f_2(\mathbf{x}) d\mathbf{x}$$

$$P(\text{misclassification as } \pi_2) = P(\mathbf{X} \in R_2 \cap \pi_1) = P(\mathbf{X} \in R_2 | \pi_1)P(\pi_1) = p_1 \int_{R_2} f_1(\mathbf{x}) d\mathbf{x}$$

Note the inclusion of prior probabilities based on relationship between joint and conditional probabilities.

The costs of misclassification can be defined by a cost matrix,

Classification Based on X	True group membership	
	π_1	π_2
$X \in R_1$	0	$c(1 2)$
$X \in R_2$	$c(2 1)$	0

1

We then define the Expected Cost of Misclassification,

2

$$\begin{aligned}
 EMC &= c(2|1)P(2|1)p_1 + c(1|2)P(1|2)p_2 \\
 &= c(2|1)p_1 \int_{R_2} f_1(x)dx + c(1|2)p_2 \int_{R_1} f_2(x)dx \\
 &= c(2|1)p_1 \left[1 - \int_{R_1} f_1(x)dx\right] + c(1|2)p_2 \int_{R_1} f_2(x)dx \\
 &= p_1 c(2|1) + \int_{R_1} \{p_2 c(1|2)f_2(x) - p_1 c(2|1)f_1(x)\}dx
 \end{aligned}$$

Manipulation so that we work with integral in same region

This shows that the EMC will be a minimum if we choose the partition R_1 such that

3

$$\int_{R_1} \{p_2 c(1|2)f_2(x) - p_1 c(2|1)f_1(x)\}dx$$

is as small as possible, or equivalently,

$$\begin{aligned}
 &x \in R_1 \\
 &p_2 c(1|2)f_2(x) - p_1 c(2|1)f_1(x) \leq 0 \\
 &p_2 c(1|2)f_2(x) \leq p_1 c(2|1)f_1(x) \\
 &\frac{f_1(x)}{f_2(x)} \geq \left(\frac{c(1|2)}{c(2|1)}\right) \left(\frac{p_2}{p_1}\right)
 \end{aligned}$$

$$\begin{aligned}
 &x \in R_2 \\
 &p_2 c(1|2)f_2(x) - p_1 c(2|1)f_1(x) > 0 \\
 &p_2 c(1|2)f_2(x) \geq p_1 c(2|1)f_1(x) \\
 &\frac{f_1(x)}{f_2(x)} < \left(\frac{c(1|2)}{c(2|1)}\right) \left(\frac{p_2}{p_1}\right)
 \end{aligned}$$

Example:

4

Suppose $c(2|1) = 5$ and $c(1|2) = 10$. Assume that 20% of all observations for which measurements x are recorded belong to π_2 . Derive the classification regions, R_1 and R_2 .

$$\begin{aligned}
 R_1: \frac{f_1(x)}{f_2(x)} &\geq \left(\frac{10}{5}\right) \left(\frac{0.2}{0.8}\right) = 0.5 \\
 R_2: \frac{f_1(x)}{f_2(x)} &< \left(\frac{10}{5}\right) \left(\frac{0.2}{0.8}\right) = 0.5
 \end{aligned}$$

Now assume that we evaluate the density functions for a new observation, x_0 , such that $f_1(x_0) = 0.3$ and $f_2(x_0) = 0.4$, do we classify the new observation to π_1 or π_2 ?

$$\frac{f_1(x_0)}{f_2(x_0)} = \frac{0.3}{0.4} = 0.75 \geq 0.5$$

So $x_0 \in R_1$ and the new observation belongs to π_1 .

Classification with two Multivariate Normal distributions

If $f_1(\mathbf{x})$ and $f_2(\mathbf{x})$ are multivariate normal densities with means $\boldsymbol{\mu}_1$ and $\boldsymbol{\mu}_2$ and a common covariance matrix $\boldsymbol{\Sigma}$ then

$$\frac{f_1(\mathbf{x})}{f_2(\mathbf{x})} = \frac{(2\pi)^{-\frac{p}{2}} |\boldsymbol{\Sigma}|^{-\frac{1}{2}} \exp[-1/2 (\mathbf{x} - \boldsymbol{\mu}_1)' \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}_1)]}{(2\pi)^{-\frac{p}{2}} |\boldsymbol{\Sigma}|^{-\frac{1}{2}} \exp[-1/2 (\mathbf{x} - \boldsymbol{\mu}_2)' \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}_2)]}$$

Using $\exp(A)/\exp(B) = \exp(A-B)$

$$= \exp[-1/2 \{ (\mathbf{x} - \boldsymbol{\mu}_1)' \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}_1) - (\mathbf{x} - \boldsymbol{\mu}_2)' \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}_2) \}]$$

Taking logs to
get rid of exp
on RHS

$$\log \left(\frac{f_1(\mathbf{x})}{f_2(\mathbf{x})} \right) = -1/2 \{ \mathbf{x}' \boldsymbol{\Sigma}^{-1} \mathbf{x} - 2\boldsymbol{\mu}_1' \boldsymbol{\Sigma}^{-1} \mathbf{x} + \boldsymbol{\mu}_1' \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_1 - [\mathbf{x}' \boldsymbol{\Sigma}^{-1} \mathbf{x} - 2\boldsymbol{\mu}_2' \boldsymbol{\Sigma}^{-1} \mathbf{x} + \boldsymbol{\mu}_2' \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_2] \}$$

$$= -1/2 \{ -2\boldsymbol{\mu}_1' \boldsymbol{\Sigma}^{-1} \mathbf{x} + \boldsymbol{\mu}_1' \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_1 + \boldsymbol{\mu}_2' \boldsymbol{\Sigma}^{-1} \mathbf{x} - \boldsymbol{\mu}_2' \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_2 \}$$

Multiplying out and
cancelling common
terms.

$$= -1/2 \{ -2(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)' \boldsymbol{\Sigma}^{-1} \mathbf{x} + \boldsymbol{\mu}_1' \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_1 + \boldsymbol{\mu}_2' \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_1 - \boldsymbol{\mu}_2' \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_1 - \boldsymbol{\mu}_2' \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_2 \}$$

$$= -1/2 \{ -2(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)' \boldsymbol{\Sigma}^{-1} \mathbf{x} + (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)' \boldsymbol{\Sigma}^{-1} (\boldsymbol{\mu}_1 + \boldsymbol{\mu}_2) \}$$

Factorising and
adding and
subtracting
 $\boldsymbol{\mu}_2' \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_1$ because
then we have a
nice quadratic term
in the next step

$$\rightarrow R_1: (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)' \boldsymbol{\Sigma}^{-1} \mathbf{x} - 1/2 (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)' \boldsymbol{\Sigma}^{-1} (\boldsymbol{\mu}_1 + \boldsymbol{\mu}_2) \geq \log \left[\left(\frac{c(1|2)}{c(2|1)} \right) \left(\frac{p_2}{p_1} \right) \right]$$

$$\rightarrow R_2: (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)' \boldsymbol{\Sigma}^{-1} \mathbf{x} - 1/2 (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)' \boldsymbol{\Sigma}^{-1} (\boldsymbol{\mu}_1 + \boldsymbol{\mu}_2) < \log \left[\left(\frac{c(1|2)}{c(2|1)} \right) \left(\frac{p_2}{p_1} \right) \right]$$

Discrimination and Classification from sample quantities

We estimate $\boldsymbol{\mu}_1, \boldsymbol{\mu}_2$ and $\boldsymbol{\Sigma}$ from a sample.

Assume we have n_1 observations of the multivariate normal random variable $\mathbf{X}' = [\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_p]$ from π_1 and n_2 from π_2 . Then

$$\mathbf{X}_{1(n_1 \times p)} = \begin{bmatrix} \mathbf{x}'_{11} \\ \mathbf{x}'_{12} \\ \vdots \\ \mathbf{x}'_{1n_1} \end{bmatrix} \text{ and } \mathbf{X}_{2(n_2 \times p)} = \begin{bmatrix} \mathbf{x}'_{21} \\ \mathbf{x}'_{22} \\ \vdots \\ \mathbf{x}'_{2n_2} \end{bmatrix} \text{ and}$$

$$\begin{aligned} \bar{\mathbf{x}}_{1(p \times 1)} &= 1/n_1 \sum_{j=1}^{n_1} \mathbf{x}_{1j} & \mathbf{S}_{1(p \times p)} &= 1/(n_1 - 1) \sum_{j=1}^{n_1} (\mathbf{x}_{1j} - \bar{\mathbf{x}}_1)(\mathbf{x}_{1j} - \bar{\mathbf{x}}_1)' \\ \bar{\mathbf{x}}_{2(p \times 1)} &= 1/n_2 \sum_{j=1}^{n_2} \mathbf{x}_{2j} & \mathbf{S}_{2(p \times p)} &= 1/(n_2 - 1) \sum_{j=1}^{n_2} (\mathbf{x}_{2j} - \bar{\mathbf{x}}_2)(\mathbf{x}_{2j} - \bar{\mathbf{x}}_2)' \end{aligned}$$

Since we assume a common $\boldsymbol{\Sigma}$, we need to calculate a pooled $\mathbf{S}_p$,

$$\mathbf{S}_p = \frac{n_1 - 1}{(n_1 - 1) + (n_2 - 1)} \mathbf{S}_1 + \frac{n_2 - 1}{(n_1 - 1) + (n_2 - 1)} \mathbf{S}_2.$$

Thus the estimated ECM rule for two normal populations will be to allocate $\mathbf{x}_0$ to π_1 if

$$(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)' \mathbf{S}_p^{-1} \mathbf{x} - 1/2(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)' \mathbf{S}_p^{-1} (\bar{\mathbf{x}}_1 + \bar{\mathbf{x}}_2) \geq \log \left[\left(\frac{c(1|2)}{c(2|1)} \right) \left(\frac{p_2}{p_1} \right) \right].$$

In the case of equal costs equal prior probabilities, $\left(\frac{c(1|2)}{c(2|1)} \right) \left(\frac{p_2}{p_1} \right) = 1$ and $\log 1 = 0$.

With equal costs and equal prior probabilities, we have zero on RHS.

From

$$(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)' \mathbf{S}_p^{-1} \mathbf{x} - 1/2(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)' \mathbf{S}_p^{-1} (\bar{\mathbf{x}}_1 + \bar{\mathbf{x}}_2) \geq 0,$$

the ECM rule for two normal populations amounts to comparing the scalar variable,

$$\hat{y} = (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)' \mathbf{S}_p^{-1} \mathbf{x} = \hat{\mathbf{a}}' \mathbf{x}$$

evaluated at $\mathbf{x}_0$ with the number

$$m = 1/2(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)' \mathbf{S}_p^{-1} (\bar{\mathbf{x}}_1 + \bar{\mathbf{x}}_2) = 1/2(\bar{y}_1 + \bar{y}_2)$$

where

$$\bar{y}_1 = (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)' \mathbf{S}_p^{-1} \bar{\mathbf{x}}_1 = \hat{\mathbf{a}}' \bar{\mathbf{x}}_1 \text{ and } \bar{y}_2 = (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)' \mathbf{S}_p^{-1} \bar{\mathbf{x}}_2 = \hat{\mathbf{a}}' \bar{\mathbf{x}}_2.$$

And this leads to a discriminant function, $\hat{y}$ and a cut-off value m

m is obtained by evaluating discriminant function using two sample group means and taking average.

This is equivalent to creating two univariate populations for the y values by taking appropriate linear combinations of the observations from the populations π_1 and π_2 and assigning a new observation, x_0 to π_1 or π_2 depending upon whether $\hat{y} = \hat{\mathbf{a}}' \mathbf{x}$ falls to the right or the left of the midpoint, $\hat{m}$ between the two univariate means $\bar{y}_1$ and $\bar{y}_2$.

Example

We are concerned about identifying hemophilia carriers based on two predictor variables.

We have sample from two populations,

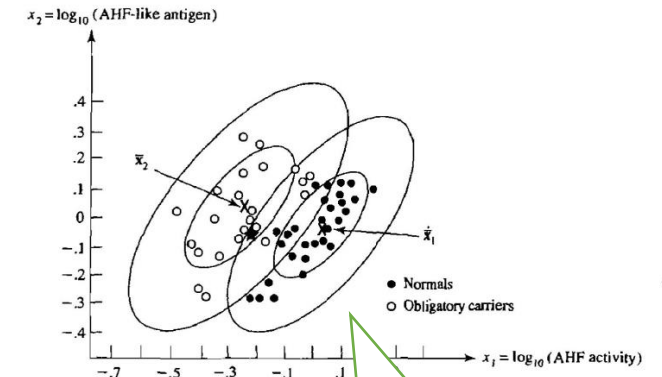
$n_1 = 30$ normal women who do not carry the hemophilia gene

$n_2 = 22$ obligatory carriers = women who do carry the hemophilia gene

and measurements on two predictor variables,

$X_1 = \log(\text{AHF activity})$ and $X_2 = \log(\text{AHF-like antigen})$

The pairs of observations on the two predictor variables for the two groups are plotted →



We have the following information for the sample,

$$\bar{x}_1 = \begin{bmatrix} -0.0065 \\ -0.0390 \end{bmatrix} \text{ and } \bar{x}_2 = \begin{bmatrix} -0.2483 \\ 0.0262 \end{bmatrix} \text{ and } S_p^{-1} = \begin{bmatrix} 131.158 & -90.423 \\ -90.423 & 108.147 \end{bmatrix}.$$

The equal costs and equal priors discriminant function is

$$\hat{y} = \hat{a}'x = (\bar{x}_1 - \bar{x}_2)'S_p^{-1}x = [0.2418 \quad -0.0652] \begin{bmatrix} 131.158 & -90.423 \\ -90.423 & 108.147 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 37.61x_1 - 28.92x_2.$$

Now we need to calculate the two univariate y -values,

$$\bar{y}_1 = \hat{a}'\bar{x}_1 = [37.61 \quad -28.92] \begin{bmatrix} -0.0065 \\ -0.0390 \end{bmatrix} = 0.88 \text{ and } \bar{y}_2 = \hat{a}'\bar{x}_2 = [37.61 \quad -28.92] \begin{bmatrix} -0.2483 \\ 0.0262 \end{bmatrix} = -10.10$$

and the midpoint between these means is $\hat{m} = 1/2(\bar{y}_1 + \bar{y}_2) = \frac{1}{2}(0.88 - 10.10) = -4.61$.

Should a woman who measures $x_1 = -0.210$ and $x_2 = -0.044$ be classified as π_1 (normal) or as π_2 (a carrier)?

We need to calculate $\hat{y}_0 = \hat{a}'x_0 = [37.61 \quad -28.92] \begin{bmatrix} -0.210 \\ -0.044 \end{bmatrix} = -6.62 < -4.61$

Which classifies the woman of population π_2 , a carrier.

Shows normal ellipses
and separation
between two groups

Fisher's Linear Discriminant Analysis

An ANOVA
type
approach.

(making no distributional assumptions but assuming a common covariance matrix.)

The general idea is to transform multivariate observations x to univariate observations y (by taking linear combinations) such that the y 's derived from populations π_1 and π_2 are separated as much as possible. We measure this separation in terms of standard deviation units estimated by the average of the within-sample standard deviations corresponding to the two populations.

This equates to maximizing the ratio $\frac{\text{squared distance between sample means of } y}{\text{pooled sample variance of } Y} = \frac{\text{Between group variance}}{\text{Within group variance}} = \frac{a'Ba}{a'Wa}$

where $B_{p \times p} = \sum_{i=1}^g (\bar{x}_{i.} - \bar{x})(\bar{x}_{i.} - \bar{x})'$ and $W_{p \times p} = \sum_{i=1}^g \sum_{j=1}^{n_i} (x_{ij} - \bar{x}_{i.})(x_{ij} - \bar{x}_{i.})'$.

We have already shown in previous chapters that the value of a for which this ratio is a maximum correspond to a being an eigenvector of $W^{-1}B$ associated with the largest eigenvalue.

It can also be shown that the Fisher LDA solution is equivalent to the minimization of the EMC under the equal misclassification costs, equal prior probabilities assumption.

So both approaches lead to the following allocation rule:

Allocate x_0 to π_1 if $\hat{y}_0 = (\bar{x}_1 - \bar{x}_2)'S^{-1}x_0 \geq 1/2 (\bar{x}_1 - \bar{x}_2)'S_p^{-1}(\bar{x}_1 + \bar{x}_2)$

Example

Back to the hemophilia example, we calculated the discriminant function to be

$$\hat{y} = \hat{a}'x = (\bar{x}_1 - \bar{x}_2)'S_p^{-1}x = 37.61x_1 - 28.92x_2$$

This is also Fisher's linear discriminant function and the maximum separation in the samples is

$$\begin{aligned} D^2 &= (\bar{x}_1 - \bar{x}_2)'S_p^{-1}(\bar{x}_1 - \bar{x}_2) \\ &= [0.2418 \quad -0.0652] \begin{bmatrix} 131.58 & -90.423 \\ -90.423 & 108.147 \end{bmatrix} \begin{bmatrix} 0.2418 \\ -0.0652 \end{bmatrix} = 10.98 \end{aligned}$$

To allocate a new observation to one of 2 populations using Fisher's Discriminant function,

$$\text{Evaluate discriminant function at } x_0, \hat{y}_0 = [37.61 \quad -28.92] \begin{bmatrix} -0.210 \\ -0.044 \end{bmatrix} = -6.62$$

$$\text{and compare to } \hat{m} = 1/2(\bar{y}_1 + \bar{y}_2) = \frac{1}{2}(\bar{x}_1 - \bar{x}_2)'S_p^{-1}(\bar{x}_1 + \bar{x}_2) = \frac{1}{2}(0.88 - 10.10) = -4.61.$$

Since $-6.62 < -4.61$, allocate to π_2 .

Quadratic Discriminant Analysis

When $\Sigma_1 \neq \Sigma_2$ the terms in the density ratio $\frac{f_1(x)}{f_2(x)}$ involving $|\Sigma_i|^{-\frac{1}{2}}$ do not cancel and the quadratic forms in the exponents of $f_1(x)$ and $f_2(x)$ do not combine:

$$\begin{aligned}\frac{f_1(x)}{f_2(x)} &= \frac{(2\pi)^{-\frac{p}{2}} |\Sigma_1|^{-\frac{1}{2}} \exp[-1/2(x - \mu_1)' \Sigma_1^{-1} (x - \mu_1)]}{(2\pi)^{-\frac{p}{2}} |\Sigma_2|^{-\frac{1}{2}} \exp[-1/2(x - \mu_2)' \Sigma_2^{-1} (x - \mu_2)]} \\&= \left(\frac{|\Sigma_1|}{|\Sigma_2|} \right)^{-1/2} \exp\left[-\frac{1}{2} \{x' \Sigma_1^{-1} x - 2\mu_1' \Sigma_1^{-1} x + \mu_1' \Sigma_1^{-1} \mu_1 - [x' \Sigma_2^{-1} x - 2\mu_2' \Sigma_2^{-1} x + \mu_2' \Sigma_2^{-1} \mu_2]\}\right] \\&= \left(\frac{|\Sigma_1|}{|\Sigma_2|} \right)^{-1/2} \exp\left[-\frac{1}{2} \{x' (\Sigma_1^{-1} - \Sigma_2^{-1}) x - 2(\mu_1' \Sigma_1^{-1} - \mu_2' \Sigma_2^{-1}) x + \mu_1' \Sigma_1^{-1} \mu_1 - \mu_2' \Sigma_2^{-1} \mu_2\}\right]\end{aligned}$$

So the partition becomes:

$$\begin{aligned}R_1: & \left(-\frac{1}{2} x' (\Sigma_1^{-1} - \Sigma_2^{-1}) x + (\mu_1' \Sigma_1^{-1} - \mu_2' \Sigma_2^{-1}) x - k \right) \geq \log \left[\left(\frac{c(1|2)}{c(2|1)} \right) \left(\frac{p_2}{p_1} \right) \right] \\R_2: & \left(-\frac{1}{2} x' (\Sigma_1^{-1} - \Sigma_2^{-1}) x + (\mu_1' \Sigma_1^{-1} - \mu_2' \Sigma_2^{-1}) x - k \right) < \log \left[\left(\frac{c(1|2)}{c(2|1)} \right) \left(\frac{p_2}{p_1} \right) \right]\end{aligned}$$

Where

$$k = \frac{1}{2} \log \left(\frac{|\Sigma_1|}{|\Sigma_2|} \right) + \frac{1}{2} (\mu_1' \Sigma_1^{-1} \mu_1 - \mu_2' \Sigma_2^{-1} \mu_2)$$

Due to lack of cancellation of $|\Sigma_i|^{-\frac{1}{2}}$, maths a little more complicated leaving us with an extra term.

Evaluating Classification Functions

Based on minimizing the "optimum error rate", $OER = p_1 \int_{R_2} f_1(x)dx + p_2 \int_{R_1} f_2(x)dx$, of which the sample equivalent is the "apparent error rate" (APER) that can be calculated from the "confusion matrix":

Classification Based on X	True Group membership	
	π_1	π_2
$X \in R_1$	n_{1C}	n_{1M}
$X \in R_2$	n_{2M}	n_{2C}
	n_1	n_2

where n_{1C} = number of π_1 items correctly classified as π_1 items, n_{1M} = number of π_1 items misclassified as π_2 items, and similarly for π_2 items.

The apparent error rate is then $APER = \frac{n_{1M} + n_{2M}}{n_1 + n_2}$ = the proportion of items in the training set that are misclassified.

BUT it underestimates the true error rate because the data used to build the classification function are also used to evaluate it.

The classification should not be tested on the same sample that was used to derive the rule. So we use a kind of cross-validation based on what is called "Lachenbruch's holdout procedure".

Classification of several populations

If we have more than 2 populations, say g populations, we can extend expected misclassification cost definition in the following way:

The conditional expected cost of misclassifying an $\mathbf{X}$ from π_1 into π_2 or π_3 , or any of the other π_i $i = 2, \dots, g$ is

$$ECM(1) = P(2|1)c(2|1) + P(3|1)c(3|1) + \dots + P(g|1)c(g|1) = \sum_{k=2}^g P(k|1)c(k|1).$$

The conditional expected cost occurs with prior probability p_1 , the prevalence of π_1 . Similarly for $ECM(2)$ to $ECM(g)$.

The overall expected cost of misclassification is

$$ECM = p_1 ECM(1) + p_2 ECM(2) + \dots + p_g ECM(g) = \sum_{i=1}^g p_i \left(\sum_{i=1(i \neq k)}^g P(k|i)c(k|i) \right),$$

which should be minimised by choosing mutually exclusive and exhaustive regions $R_1, R_2, \dots, R_g$.

The classification regions that minimize the ECM are defined by allocating $\mathbf{x}$ to that population, $\pi_1, \pi_2, \dots, \pi_g$, for which

$$\sum_{i=1(i \neq k)}^g p_i f_i(\mathbf{x}) c(k|i) \text{ is smallest.}$$

If all misclassification costs are equal,

The MINIMUM CLASSIFICATION RULE WITH EQUAL MISCLASSIFICATION COSTS is to

allocate $\mathbf{x}_0$ to π_k if $p_k f_k(\mathbf{x}_0) > p_i f_i(\mathbf{x}_0)$ for all $i \neq k$

or, equivalently,

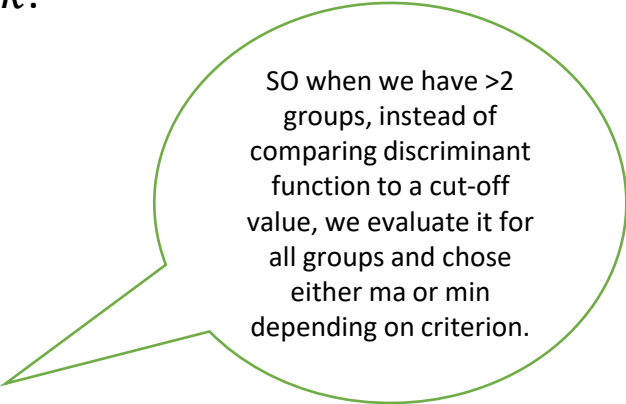
allocate $\mathbf{x}_0$ to π_k if $\ln p_k f_k(\mathbf{x}_0) > \ln p_i f_i(\mathbf{x}_0)$ for all $i \neq k$.

It can be shown that this is equivalent to calculating the discriminant score,

$$\hat{d}_i(x) = \bar{x}_i' S_p^{-1} x - \frac{1}{2} \bar{x}_i' S_p^{-1} \bar{x}_i + \ln p_i$$

and using the rule that allocates x to π_k if

$$\hat{d}_k(x) = \max(\hat{d}_1(x), \hat{d}_2(x), \dots, \hat{d}_p(x)).$$



SO when we have >2 groups, instead of comparing discriminant function to a cut-off value, we evaluate it for all groups and chose either ma or min depending on criterion.

Alternatively, define the discriminant score in terms of squared distances,

$$D_i^2(x) = (x - \bar{x}_i)' S_p^{-1} (x - \bar{x}_i)$$

and allocate x to π_i for which $-\frac{1}{2} D_i^2(x) + \ln p$ is the largest (and assigns x to the 'closest' population).

Can also use Fisher's method to classify observation into one of multiple groups:

Based on finding coefficients a that maximizes $\frac{a'Ba}{a'Wa}$

where $B_{p \times p} = \sum_{i=1}^g n_i (\bar{x}_{i.} - \bar{x})(\bar{x}_{i.} - \bar{x})'$ and $W_{p \times p} = \sum_{i=1}^g \sum_{j=1}^{n_i} (x_{ij} - \bar{x}_{i.})(x_{ij} - \bar{x}_{i.})' = \sum_{i=1}^g (n_i - 1)S_i$.

Also $W / (n_1 + n_2 + \dots + n_g - g) = S_{pooled}$.

The first sample discriminant function is $\hat{a}'_1 x$ where $\hat{a}_1 = \hat{e}_1$, the eigenvector of $W^{-1}B$ associated with the largest eigenvalue, $\hat{\lambda}_1$ and the k^{th} discriminant function is $\hat{a}'_k x = \hat{e}'_k x$ with $\hat{a}'_i S_p \hat{a}_i = 1$.

To use the sample discriminants for classification we will allocate x to π_k if

$$\sum_{j=1}^r (\hat{y}_j - \bar{y}_{kj})^2 = \sum_{j=1}^r [\hat{a}'_j (x - \bar{x}_k)]^2 \leq \sum_{j=1}^r [\hat{a}'_j (x - \bar{x}_i)]^2 \text{ for } i \neq k.$$

To summarise:

1. Classification of observation into one of two classes based on ECM which leads to $\frac{f_1(x)}{f_2(x)} \geq \left(\frac{c(1|2)}{c(2|1)}\right) \left(\frac{p_2}{p_1}\right)$.

All you
need on
one page!

2. Substitute MVNs for $f_1(x)$ and $f_2(x) \rightarrow R_1: (\mu_1 - \mu_2)' \Sigma^{-1} x - 1/2(\mu_1 - \mu_2)' \Sigma^{-1}(\mu_1 + \mu_2) \geq \log \left[\left(\frac{c(1|2)}{c(2|1)}\right) \left(\frac{p_2}{p_1}\right) \right]$.

3. Replace with sample estimates $\rightarrow$ allocate x_0 to π_1 if $\underbrace{(\bar{x}_1 - \bar{x}_2)' S_p^{-1} x_0}_{y_0} - \frac{1}{2} \underbrace{(\bar{x}_1 - \bar{x}_2)' S_p^{-1} (\bar{x}_1 + \bar{x}_2)}_m \geq \log \left[\left(\frac{c(1|2)}{c(2|1)}\right) \left(\frac{p_2}{p_1}\right) \right]$.

4. Note that for equal cost and equal prior probabilities the RHS simplifies to 0.

5. Fisher's LDA based on maximizing $\frac{\text{Between group variance}}{\text{Within group variance}} = \frac{a' B a}{a' W a}$ where $\mathbf{B}_{p \times p} = \sum_{i=1}^g n_i (\bar{x}_{i.} - \bar{x})(\bar{x}_{i.} - \bar{x})'$ and $\mathbf{W}_{p \times p} = \sum_{i=1}^g \sum_{j=1}^{n_i} (x_{ij} - \bar{x}_{i.})(x_{ij} - \bar{x}_{i.})'$ and a is the eigenvector of $W^{-1}B$ associated with the largest eigenvalue. Can be shown to be equivalent to 3.

5. Quadratic DA when two covariance matrices are not equal ...

6. Classification into multiple groups : Either allocate x to π_k if $\hat{d}_k(x) = \max(\hat{d}_1(x), \hat{d}_2(x), \dots, \hat{d}_p(x))$, where $\hat{d}_i(x) = \bar{x}_i' S_p^{-1} x - \frac{1}{2} \bar{x}_i' S_p^{-1} \bar{x}_i + \ln p_i$ or calculate $D_i^2(x) = (x - \bar{x}_i)' S_p^{-1} (x - \bar{x}_i)$ and allocate x to π_i for which $-\frac{1}{2} D_i^2(x) + \ln p$ is the largest or use Fisher's method $\sum_{j=1}^r (\hat{y}_j - \bar{y}_{kj})^2 = \sum_{j=1}^r [\hat{a}_j'(x - \bar{x}_k)]^2 \leq \sum_{j=1}^r [\hat{a}_j'(x - \bar{x}_i)]^2$ for $i \neq k$.